

Class Notes 10

Waveguides in Cartesian Coordinates

Balanis Chapt 8 Sections 8-1 through 8.2.3. Before we associated the magnetic $\mathbf{A}$ and electric $\mathbf{F}$ vector potentials with the electric $\mathbf{J}$ and magnetic $\mathbf{M}$ currents sources. Now we will look at an entirely different theory where the magnetic and electric vector potentials are associated with a particular direction in space with modes TE or TM to that direction.

Construction of Solutions in a Source Free Region

$\nabla \times \mathbf{E} = -j\omega\mu\mathbf{H}$	(1)	An arbitrary field in a source-free region can be expressed as a TE field, a TM field or a sum of a TE and a TM field to some direction $\hat{\mathbf{u}}$
$\nabla \times \mathbf{H} = j\omega\varepsilon\mathbf{E}$	(2)	
$\nabla \cdot \mathbf{H} = 0$	(3)	
$\nabla \cdot \mathbf{E} = 0$	(4)	

TM Solutions

$$\nabla^2 \mathbf{A} + k^2 \mathbf{A} = 0 \quad (5a)$$

$$\mathbf{A} = \psi^{TM} \hat{\mathbf{u}} \quad (6a)$$

$$\mathbf{E} = \frac{-j\omega}{k^2} (k^2 \mathbf{A} + \nabla \nabla \cdot \mathbf{A}) \quad (7a)$$

$$\mathbf{H} = \frac{1}{\mu} \nabla \times \mathbf{A} \quad (8a)$$

TE Solutions

$$\nabla^2 \mathbf{F} + k^2 \mathbf{F} = 0 \quad (5b)$$

$$\mathbf{F} = \psi^{TE} \hat{\mathbf{u}} \quad (6b)$$

$$\mathbf{H} = \frac{-j\omega}{k^2} (k^2 \mathbf{F} + \nabla \nabla \cdot \mathbf{F}) \quad (7b)$$

$$\mathbf{E} = -\frac{1}{\varepsilon} \nabla \times \mathbf{F} \quad (8b)$$

If the actual fields are neither TE nor TM to some direction $\hat{\mathbf{u}}$, but a hybrid combination of the two, then

$$\mathbf{E} = \frac{-j\omega}{k^2} (k^2 \mathbf{A} + \nabla \nabla \cdot \mathbf{A}) - \frac{1}{\varepsilon} \nabla \times \mathbf{F}$$

$$\mathbf{H} = \frac{-j\omega}{k^2} (k^2 \mathbf{F} + \nabla \nabla \cdot \mathbf{F}) + \frac{1}{\mu} \nabla \times \mathbf{A}$$

Solutions in terms of scalar potentials in rectangular coordinates

TM case

$$\mathbf{H} = \frac{1}{\mu} \nabla \times \mathbf{A} \quad (1)$$

$$\mathbf{E} = -\frac{j\omega}{k^2} \left(k^2 \mathbf{A} + \nabla \nabla \cdot \mathbf{A} \right) \quad (2)$$

$$\nabla^2 \mathbf{A} + k^2 \mathbf{A} = 0 \quad (3)$$

Let

$$\mathbf{A} = \psi^{TM} \hat{\mathbf{u}} \quad (4)$$

The unit vector $\hat{\mathbf{u}}$ is fixed, i.e. not a function of position (x, y, z) , so

$$\nabla \times \mathbf{A} = \nabla \times (\psi^{TM} \hat{\mathbf{u}}) = \left(\cancel{\psi^{TM} \nabla \times \hat{\mathbf{u}}}^0 - \hat{\mathbf{u}} \times \nabla \psi^{TM} \right) \quad (5)$$

or
$$\mathbf{H} = \frac{1}{\mu} \nabla \psi^{TM} \times \hat{\mathbf{u}} \quad (6)$$

Notice that $\mathbf{H}$ is $\perp$ to $\hat{\mathbf{u}}$ so it does not have a component in the direction of $\hat{\mathbf{u}}$. The field is therefore transverse magnetic to $\hat{\mathbf{u}}$.

Substitute (4) into (2)

$$\mathbf{E} = \frac{-j\omega}{k^2} \left(k^2 \psi^{TM} \hat{\mathbf{u}} + \nabla \nabla \cdot (\psi^{TM} \hat{\mathbf{u}}) \right)$$

but
$$\nabla \cdot (\psi \hat{\mathbf{u}}) = \psi \nabla \cdot \hat{\mathbf{u}} + \hat{\mathbf{u}} \cdot \nabla \psi = \hat{\mathbf{u}} \cdot \nabla \psi = \frac{\partial \psi}{\partial u}$$

so
$$\mathbf{E} = -\frac{j\omega}{k^2} \left(k^2 \psi^{TM} \hat{\mathbf{u}} + \nabla \frac{\partial \psi^{TM}}{\partial u} \right) \quad (7)$$

Substitute (4) into (3) to get

$$\nabla^2 \psi^{TM} + k^2 \psi^{TM} = 0 \quad (8)$$

TE case

Similarly let $\mathbf{F} = \psi^{TE} \hat{\mathbf{u}} \Rightarrow$

$$\mathbf{E} = -\frac{1}{\epsilon} \hat{\mathbf{u}} \times \nabla \psi^{TE} \quad (9)$$

$$\mathbf{H} = \frac{-j\omega}{k^2} \left(k^2 \psi^{TE} \hat{\mathbf{u}} + \nabla \frac{\partial \psi^{TE}}{\partial u} \right) \quad (10)$$

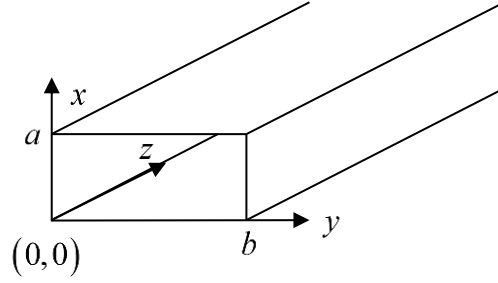
$$\nabla^2 \psi^{TE} + k^2 \psi^{TE} = 0 \quad (11)$$

TM Solutions

$$\mathbf{H}^{TM} = \mathbf{H}_{transv}^{TM} \quad (12)$$

$$\mathbf{E}^{TM} = \mathbf{E}_{transv}^{TM} + E_u^{TM} \hat{\mathbf{u}} \quad (13)$$

- Application to rectangular waveguide.



TM_z

$$\mathbf{A} = \psi \hat{\mathbf{z}}$$

$$\mathbf{H} = \frac{1}{\mu} \nabla \times \mathbf{A} = \frac{1}{\mu} \nabla \psi \times \hat{\mathbf{z}}$$

$$\mathbf{E} = -\frac{j\omega}{k^2} \left(k^2 \mathbf{A} + \nabla \nabla \cdot \mathbf{A} \right) = -\frac{j\omega}{k^2} \left(k^2 \psi \hat{\mathbf{z}} + \nabla \frac{\partial}{\partial z} \psi \right)$$

from which we get

$$H_x = \frac{1}{\mu} \frac{\partial}{\partial y} \psi \quad (14)$$

$$E_x = \frac{-j\omega}{k^2} \frac{\partial^2}{\partial x \partial z} \psi \quad (17)$$

$$H_y = -\frac{1}{\mu} \frac{\partial}{\partial x} \psi \quad (15)$$

$$E_y = -\frac{j\omega}{k^2} \frac{\partial^2}{\partial y \partial z} \psi \quad (18)$$

$$H_z = 0 \quad (16)$$

$$E_z = -\frac{j\omega}{k^2} \left(k^2 \psi + \frac{\partial^2}{\partial z^2} \psi \right) \quad (19)$$

also since from (8)

$$\nabla^2 (\psi \hat{\mathbf{z}}) + k^2 (\psi \hat{\mathbf{z}}) = 0$$

$$\Rightarrow \nabla^2 \psi + k^2 \psi = 0 \quad (20)$$

for TE_z

let $\mathbf{F} = \psi \hat{\mathbf{z}}$

so $\mathbf{E} = -\frac{1}{\epsilon} \nabla \times \mathbf{F} = -\frac{1}{\epsilon} \nabla \psi \times \hat{\mathbf{z}}$

$$\mathbf{H} = -\frac{j\omega}{k^2} \left(k^2 \mathbf{F} + \nabla \nabla \cdot \mathbf{F} \right) = -\frac{j\omega}{k^2} \left(k^2 \psi \hat{\mathbf{z}} + \nabla \frac{\partial}{\partial z} \psi \right)$$

which gives

$$H_x = \frac{-j\omega}{k^2} \frac{\partial^2}{\partial x \partial z} \psi \quad (21)$$

$$E_x = -\frac{1}{\epsilon} \frac{\partial}{\partial y} \psi \quad (24)$$

$$H_y = \frac{-j\omega}{k^2} \frac{\partial^2}{\partial y \partial z} \psi \quad (22)$$

$$E_y = \frac{1}{\epsilon} \frac{\partial}{\partial x} \psi \quad (25)$$

$$H_z = \frac{-j\omega}{k^2} \left(k^2 \psi + \frac{\partial^2}{\partial z^2} \psi \right) \quad (23)$$

$$E_z = 0 \quad (26)$$

and $\nabla^2 \psi + k^2 \psi = 0 \quad (27)$



In both TE and TM cases the field in the guide is found by solving the scalar wave equation. This is carried out by separation of variables:

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} + k^2 \right) \psi = 0 \quad (27)$$

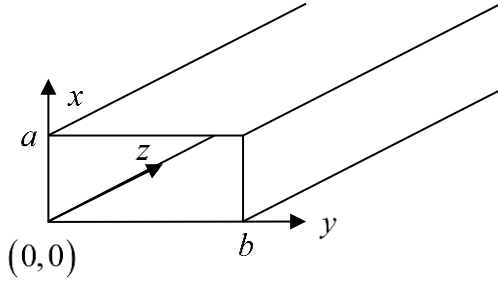
let $\psi = X(x)Y(y)Z(z) \quad (28)$

Substitute (27) into (28),

$$YZ \frac{\partial^2 X}{\partial x^2} + XZ \frac{\partial^2 Y}{\partial y^2} + XY \frac{\partial^2 Z}{\partial z^2} + k^2 XYZ = 0 \quad (29)$$

or

$$\begin{aligned} \frac{1}{X} \frac{\partial^2}{\partial x^2} X + \frac{1}{Y} \frac{\partial^2}{\partial y^2} Y + \frac{1}{Z} \frac{\partial^2}{\partial z^2} Z + k^2 &= 0 \quad (30) \\ \doteq -k_x^2 &\quad \doteq -k_y^2 \quad \doteq -k_z^2 \end{aligned}$$



from which we get the auxiliary equation
also called the dispersion relation

$$k_x^2 + k_y^2 + k_z^2 - k^2 = 0 \quad (31)$$

and

$$\begin{aligned} \left(\frac{d^2}{dx^2} + k_x^2 \right) X &= 0 \\ \left(\frac{d^2}{dy^2} + k_y^2 \right) Y &= 0 \\ \left(\frac{d^2}{dz^2} + k_z^2 \right) Z &= 0 \end{aligned} \quad (32)$$

These equations are all of the form

$$\left(\frac{d^2}{d\xi^2} + \lambda^2 \right) f = 0 \quad (33)$$

which has the solution

$$f = Ae^{-j\lambda\xi} + Be^{j\lambda\xi} \quad (34)$$

or

$$f = A' \cos(\lambda\xi) + B' \sin(\lambda\xi) \quad (35)$$

So either (34) or (35) is substituted into (28) for each of the 3 variables $X(x)$, $Y(y)$, and $Z(z)$ to get the solution. Then boundary conditions are applied.

Sometimes the field must have a particular value at a boundary but, if there is no second boundary, it must go to zero at ∞ or if there is a second boundary, which may be at a higher potential for example, it may grow. The solutions to (33) in that case are

$$f = Ae^{-\lambda\xi} + Be^{\lambda\xi} \quad (36)$$

or

$$f = A' \cosh(\lambda\xi) + B' \sinh(\lambda\xi) \quad (37)$$

Like sin and cos, sinh is an odd function about the origin and cosh is an even function with a value of one at the origin, but both go to infinity at the two ends.

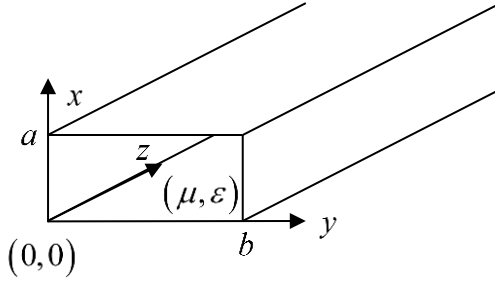
TE_z

From electric vector potential

$$\mathbf{E} = -\frac{1}{\epsilon} \nabla \times \mathbf{F} \quad (1)$$

$$\mathbf{H} = -\frac{j\omega}{k^2} (k^2 \mathbf{F} + \nabla \nabla \cdot \mathbf{F}) \quad (2)$$

Let $\mathbf{F}(\mathbf{r}) = \psi^{TE}(\mathbf{r}) \hat{\mathbf{z}}$ (3)



Then from the above equations

$$H_x = \frac{-j\omega}{k^2} \frac{\partial^2 \psi^{TE}}{\partial x \partial z} \quad (4)$$

$$E_x = -\frac{1}{\epsilon} \frac{\partial \psi^{TE}}{\partial y} \quad (7)$$

$$H_y = \frac{-j\omega}{k^2} \frac{\partial^2 \psi^{TE}}{\partial y \partial z} \quad (5)$$

$$E_y = \frac{1}{\epsilon} \frac{\partial \psi^{TE}}{\partial x} \quad (8)$$

$$H_z = \frac{-j\omega}{k^2} \left(k^2 + \frac{\partial^2}{\partial z^2} \right) \psi^{TE} \quad (6)$$

$$E_z = 0 \quad (9)$$

ψ^{TE} must satisfy $(\nabla^2 + k^2) \psi^{TE} = 0$

Assume $\psi^{TE}(\mathbf{r})$ is separable $\Rightarrow \psi^{TE}(\mathbf{r}) = X(x)Y(y)Z(z)$ (10)

The boundary conditions are:

(BC1) $E_x(x, 0, z) = E_x(x, b, z) = 0$ for $x \in (0, a)$ and all z

(BC2) $E_y(0, y, z) = E_y(a, y, z) = 0$ for $y \in (0, b)$ and all z

Substitute (10) into (7) and (8) above

$$E_x = -X(x) \frac{\partial Y(y)}{\partial y} Z(z)$$

$$E_y = \frac{\partial X(x)}{\partial x} Y(y) Z(z)$$

$$(BC1) \Rightarrow E_x = -X(x) \frac{\partial}{\partial y} Y(y) Z(z) = 0 \text{ at } y = 0 \text{ \& } y = b$$

$$\Rightarrow \frac{\partial Y(y=0)}{\partial y} = \frac{\partial Y(y=b)}{\partial y} = 0$$

$$(BC2) \Rightarrow E_y = \frac{\partial X(x)}{\partial x} Y(y) Z(z) = 0 \text{ at } x = 0 \text{ \& } x = b$$

$$\Rightarrow \frac{\partial X(x=0)}{\partial x} = \frac{\partial X(x=b)}{\partial x} = 0$$

Choose the sine, cosine form of the solution for $X(x)$:

$$X(x) = C_1 \cos(k_x x) + D_1 \sin(k_x x)$$

So from (BC1)

$$\frac{\partial X(x)}{\partial x} = C_1 k_x \sin(k_x x) - D_1 k_x \cos(k_x x)$$

$$\frac{\partial X(x=0)}{\partial x} = 0 \Rightarrow D_1 = 0$$

$$\frac{\partial X(x=a)}{\partial x} = 0 \Rightarrow \sin k_x a = 0 \Rightarrow k_x a = 0, \pi, 2\pi, 3\pi, \dots$$

$$\therefore \underline{X(x) = C_1 \cos k_x x, \quad k_x = \frac{m\pi}{a}; m = 0, 1, 2, 3, \dots}$$

Choose the sine, cosine form of the solution for $X(x)$:

$$Y(y) = C_2 \cos(k_y y) + D_2 \sin(k_y y)$$

From (BC2)

$$\underline{Y(y) = C_2 \cos k_y y, \quad k_y = \frac{n\pi}{b}; n = 0, 1, 2, 3, \dots}$$

and from $\frac{\partial^2}{\partial z^2} Z + k_z^2 Z = 0$ and no boundaries in z

$$\underline{Z = E e^{\pm j k_z z}} \quad ; \quad \pm \Rightarrow \text{either } + \text{ or } - \text{ but not both together}$$

From the auxiliary (dispersion) equation:

$$\begin{aligned} -k_x^2 - k_y^2 - k_z^2 + k^2 &= 0 \\ \Rightarrow \underline{k_z = \sqrt{k^2 - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2}} \end{aligned}$$

$\therefore$ The TE_z mode functions are:

$$\boxed{\psi_{mn}^{TEz} = A_{mn} \cos(k_x x) \cos(k_y y) e^{\pm j k_z z}}$$

where $A_{mn} = C_1 C_2 E$ with each mode coefficient distinguished by two numbers m and n . Substitute this into (4-9) to get the fields

Note: $m = 0$ or $n = 0$ is ok but not both $m = n = 0$

Rectangular waveguides

$$\psi_{mn}^{TM} = \sin(k_x x) \sin(k_y y) e^{-jk_z z}$$

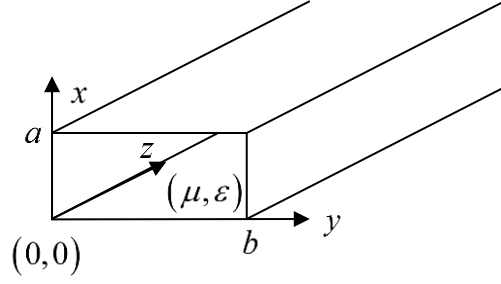
$$\psi_{mn}^{TE} = \cos(k_x x) \cos(k_y y) e^{-jk_z z}$$

In both cases

$$k_x = \frac{m\pi}{a}, k_y = \frac{n\pi}{b}$$

and

$$k_x^2 + k_y^2 + k_z^2 - k^2 = 0 \Rightarrow k_z = \sqrt{k^2 - k_x^2 - k_y^2} = \sqrt{k^2 - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2}$$



Cutoff $\equiv$ frequency at which a mode ceases to propagate ($k_z = 0$)

$$k_z = 0 = \sqrt{k_c^2 \mu \epsilon - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2}$$

$$\Rightarrow k_{c,mn} = \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}$$

$$f_{c,mn} = \frac{k_{c,mn}}{2\pi\sqrt{\mu\epsilon}} = \frac{1}{2\sqrt{\mu\epsilon}} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

$$\lambda_{c,mn} = \frac{2\pi}{k_{c,mn}} = \frac{2}{\sqrt{\left(m/a\right)^2 + \left(n/b\right)^2}}$$

Wave impedance $\equiv$ (positive) ratio of transverse electric and magnetic field components

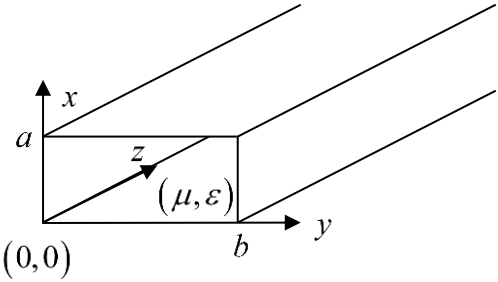
$$(Z_0)_{mn}^{TM} = \frac{E_x}{H_y} = -\frac{E_y}{H_x} = \frac{k_z}{\omega\epsilon}$$

$$(Z_0)_{mn}^{TE} = \frac{E_x}{H_y} = -\frac{E_y}{H_x} = \frac{\omega\mu}{k_z}$$

Worksheet

TE_z

$\mathbf{F}(\mathbf{r}) = \psi^{TE}(\mathbf{r}) \hat{\mathbf{z}}$



$\mathbf{H} =$

$\mathbf{E} =$

$H_x =$

$E_x =$

$H_y =$

$E_y =$

$H_z =$

$E_z =$

(BC1)

(BC2)

$\psi^{TE}(\mathbf{r}) = X(x)Y(y)Z(z)$
=

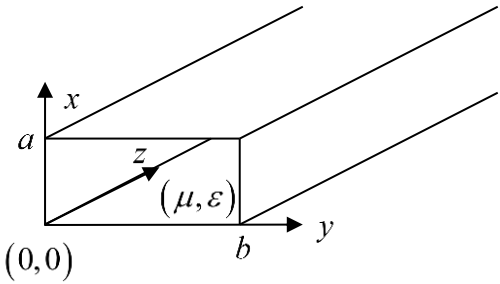
$\psi^{TE}(\mathbf{r}) =$

Worksheet

TM_z

$\mathbf{F}(\mathbf{r}) =$

$\mathbf{A}(\mathbf{r}) =$



$\mathbf{H} =$

$\mathbf{E} =$

$H_x =$

$E_x =$

$H_y =$

$E_y =$

$H_z =$

$E_z =$

(BC1)

(BC2)

$\psi^{TE}(\mathbf{r}) = X(x)Y(y)Z(z)$
=

$\psi^{TE}(\mathbf{r}) =$